VIT BHOPAL

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Course Name : INTRODUCTION TO COMPUTATIONAL CHEMISTRY

Course Code : CHY1005

School Name : SCOPE

Slot : C11 + C12 + C13

SEMESTER : WINTER SEMESTER

Faculty : Dr. Saurav Prasad

Introduction:

- Quantum chemistry studies the behavior of particles at atomic scale.
- It is governed by wave mechanics and probability.
- Mathematical models are essential to understand these systems.

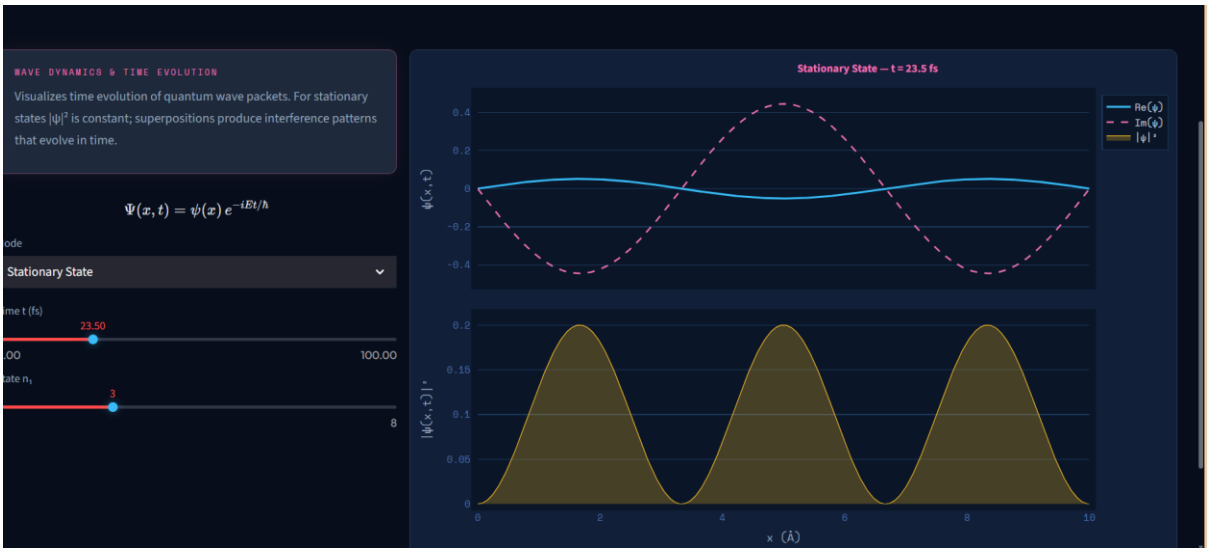
Key Equation-

Schrödinger Equation:

$\hat{H}\psi = E\psi$

Objective:

- To visualize quantum chemistry concepts using interactive simulations.
- To simplify complex equations like the Schrödinger Equation.
- To display wavefunctions, probability density, and energy levels graphically.
- To improve understanding through real-time computational models.



PROBLEM STATEMENT:

Quantum mechanics involves abstract concepts and complex mathematical equations, which are difficult to visualize and understand. Concepts like wavefunctions and uncertainty principles ($\Delta x \cdot \Delta p \geq h/2$) are not easily explained through traditional methods.

Conventional teaching lacks interactive visualization, making it hard for students to connect theory with real understanding.

Solution:

Develop a simulation-based visualization tool.

TECHNOLOGIES USED

- Python – Core programming
- Streamlit – Web interface
- Tkinter – Desktop GUI
- NumPy – Numerical computation
- Plotly & Matplotlib – Visualization
- CSS – UI styling

Used for solving equations and generating simulations

SYSTEM ARCHITECTURE:

1. User inputs parameters (n, Z, L, etc.)
2. Backend applies quantum equations
3. Computations performed using NumPy
4. Graphs generated using Plotly/Matplotlib
5. Output displayed interactively

Example Equation Used:

$\lambda = h / mv$

Flow:

User Input → Equation Processing → Visualization → Output

CORE MODULES:

- **Bohr's Model:**

$E_n = -13.6\ (Z^2 / n^2)$

- **Quantum Foundations:**

$\lambda = h / mv$

$\Delta x \cdot \Delta p \geq \hbar/2$

- **Particle in a Box:**

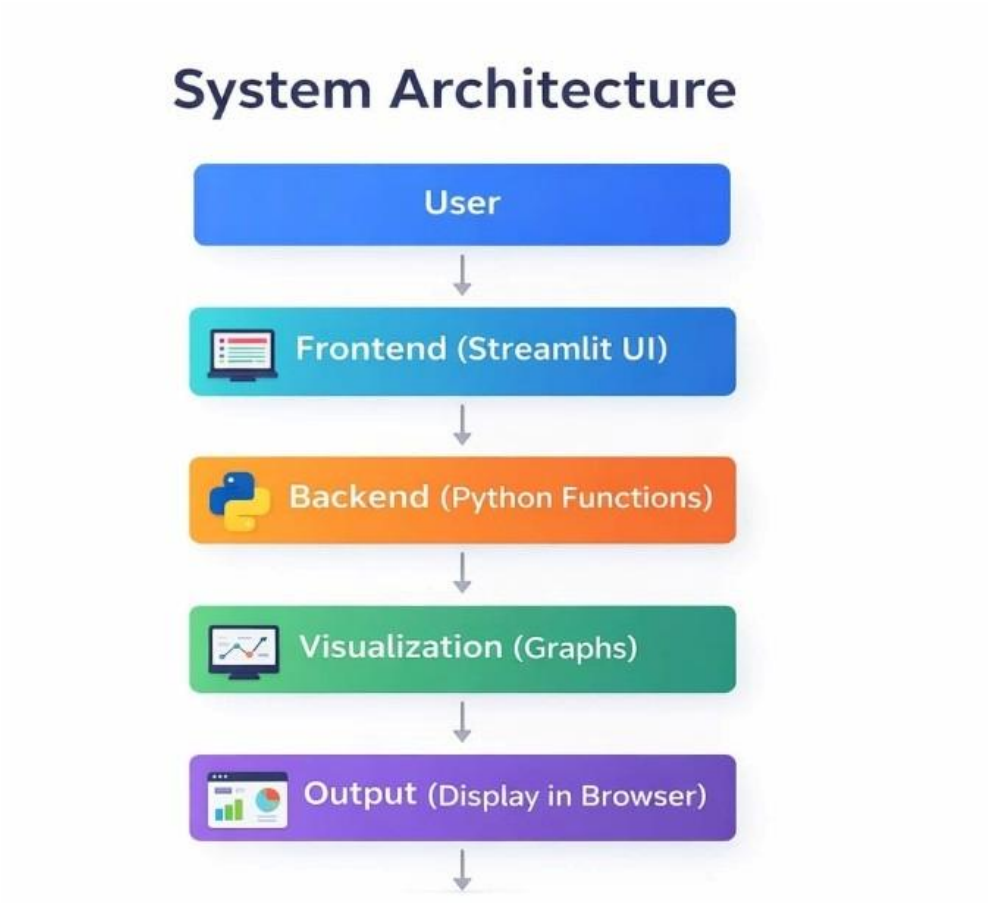
$E_n = n^2\pi^2\hbar^2 / 2mL^2$

- Wave Dynamics:

$\Psi(x,t) = \psi(x)e^{(-iEt/\hbar)}$

- 3D Orbital Lab:
  Provides interactive visualization for better understanding

  a. Helps visualize abstract quantum concepts clearly
  b. Improves understanding through real-time interaction
  c. Bridges theory with practical simulation



IMPLEMENTATION:

- Modular functions implemented for equations
- Each concept translated into computational form
- Mathematical formulas implemented using Python libraries (NumPy, Matplotlib)
- Interactive UI developed using Streamlit
- User inputs taken dynamically for real-time calculations
- Graphs generated for wavefunctions and probability density

Examples:

- bohr_energy() → uses $E_n = -13.6(Z^2/n^2)$
- pib_wavefunction() → $\psi(x) = \sqrt{(2/L)}\ \sin(n\pi x/L)$
- UI integrates mathematical models with visualization

```python
# ── PAGE HEADER ──────────────────────────────────────────────────────────
st.markdown("""
<div class="suite-header">
  <div class="header-glow"></div>
  <div class="orbit-ring r1"></div>
  <div class="orbit-ring r2"></div>
  <div class="orbit-ring r3"></div>
  <div class="header-content">
    <div class="vit-badge">⚛ VIT BHOPAL UNIVERSITY</div>
    <h1 class="suite-title">QUANTUM CHEMISTRY <span class="accent">SUITE</span></h1>
    <p class="suite-subtitle">Computational Visualization · Module 2 · v18.0</p>
    <div class="header-meta">
      <span class="meta-pill">👨‍🏫 Dr. Saurav Prasad</span>
      <span class="meta-pill">ℹ️ Slot C11+C12+C13</span>
      <span class="meta-pill">🧪 Intro to Computational Chemistry</span>
    </div>
  </div>
```

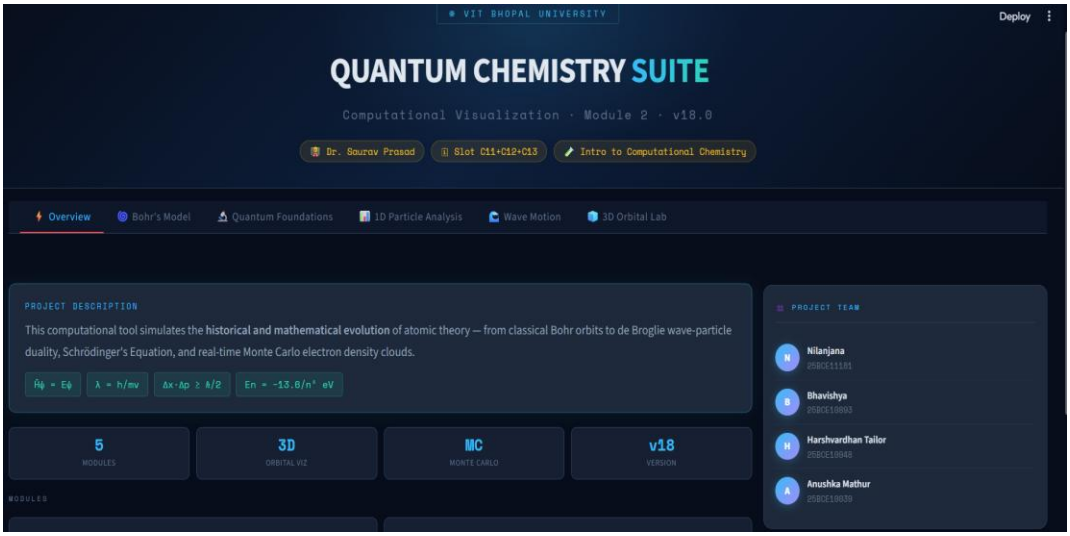
```python
# ── QUANTUM PHYSICS BACKEND ──────────────────────────────────────────────

def bohr_energy(n, Z=1):
    """En = -13.6 Z²/n²  eV"""
    return -13.6 * (Z ** 2) / (n ** 2)


def bohr_radius(n, Z=1):
    """rn = 0.529 n²/Z  Å"""
    return 0.529 * (n ** 2) / Z


def pib_wavefunction(n, L=10.0, pts=600):
    x = np.linspace(0, L, pts)
    psi = np.sqrt(2.0 / L) * np.sin(n * np.pi * x / L)
    return x, psi


def pib_energy(n, L_m, mass=9.109e-31):
    """Energy eigenvalue in eV"""
    hbar = 1.0546e-34
    return (n ** 2 * np.pi ** 2 * hbar ** 2) / (2 * mass * L_m ** 2 * 1.602e-19)


def orbital_cloud(n, num=6000):
    """Monte Carlo electron positions for hydrogen-like orbital"""
    r     = np.random.gamma(n + 2, 1.5 * n, num)
    theta = np.random.uniform(0, np.pi, num)
    phi   = np.random.uniform(0, 2 * np.pi, num)
    x = r * np.sin(theta) * np.cos(phi)
    y = r * np.sin(theta) * np.sin(phi)
    z = r * np.cos(theta)
```

OUTPUT:

- Graphs of wavefunctions ψ(x)
- Probability density plots $|\psi|^2$
- Energy level visualization

Key Concept:

Probability Density = $|\psi|^2$
- Real-time interaction improves understanding of equations


VIT BHOPAL UNIVERSITY
Deploy
QUANTUM CHEMISTRY SUITE
Computational Visualization · Module 2 · v18.0
Dr. Saurav Prasad
Slot C11+C12+C13
Intro to Computational Chemistry
Overview
Bohr's Model
Quantum Foundations
1D Particle Analysis
Wave Motion
3D Orbital Lab
PROJECT DESCRIPTION
This computational tool simulates the historical and mathematical evolution of atomic theory — from classical Bohr orbits to de Broglie wave-particle duality, Schrödinger's Equation, and real-time Monte Carlo electron density clouds.
Ĥψ = Eψ
λ = h/mv
Δx·Δp ≥ ħ/2
En = -13.6/n² eV
5
MODULES
3D
ORBITAL VIZ
MC
MONTE CARLO
v18
VERSION
MODULES
PROJECT TEAM
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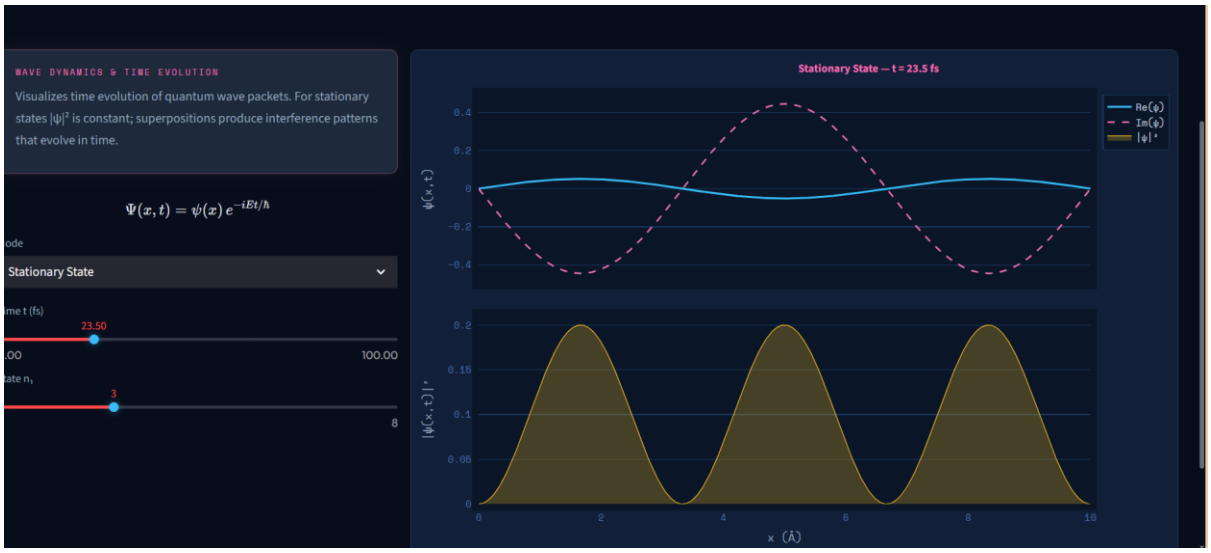
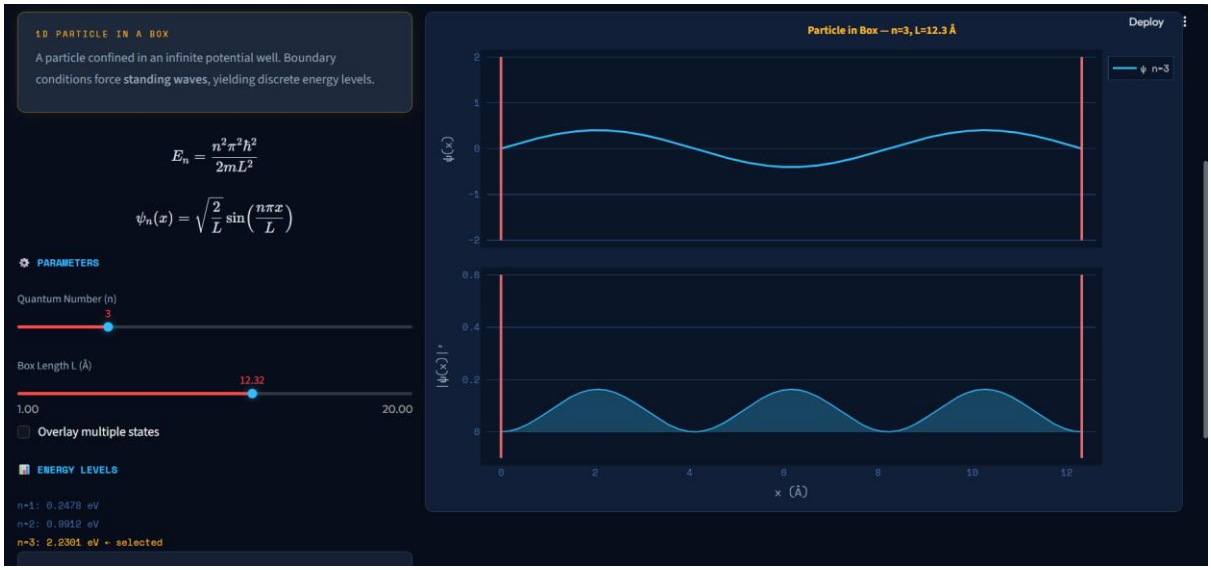
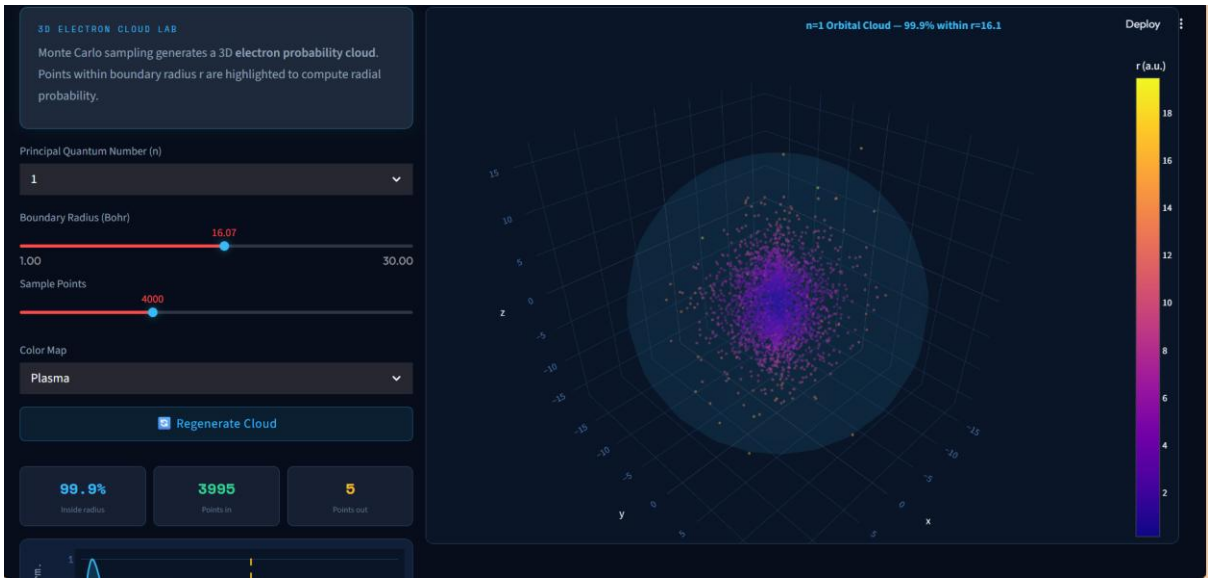
BOHR'S ATOMIC MODEL (1913)
Electrons orbit the nucleus in discrete, circular energy levels. Energy is quantized — only specific orbits n = 1, 2, 3 … are allowed.
PARAMETERS
omic Number (Z)
4
10
ow orbits up to n
4
KEY EQUATIONS
$E_n = -13.6 \frac{Z^2}{n^2}$ eV
$r_n = 0.529 \frac{n^2}{Z}$ Å
$mvr = \frac{nh}{2\pi}$
ENERGY TABLE
Energy (eV)
Radius (Å)
Bohr Model — Z=4, n up to 4
Deploy
Nucleus
n=1
e⁻ n=1
n=2
e⁻ n=2
n=3
e⁻ n=3
n=4
e⁻ n=4
y (Å)
x (Å)
Energy Level Diagram
E = 0 (Ionization)
Energy (eV)


WAVE DYNAMICS & TIME EVOLUTION
Visualizes time evolution of quantum wave packets. For stationary states |ψ|² is constant; superpositions produce interference patterns that evolve in time.
$\Psi(x,t) = \psi(x)\, e^{-iEt/\hbar}$
ode
Stationary State
me t (fs)
23.50
.00
100.00
ate n
3
8
Stationary State — t = 23.5 fs
Re(ψ)
Im(ψ)
|ψ|²
ψ(x,t)
|ψ(x,t)|²
x (Å)





Conclusion:

- Successfully integrates quantum equations into visual simulations
- Bridges gap between mathematical theory and practical understanding
- Enhances learning through interactive models

Future Scope:

- Extend the system to support multi-electron atoms and complex quantum systems

- Incorporate advanced quantum models like Schrödinger-based simulations in 3D
- Integrate AI/ML techniques for predictive analysis and faster computations
- Enhance visualization with more interactive and real-time 3D simulations
- Improve computational efficiency and performance optimization
- Develop a web-based or cloud platform for wider accessibility
- Add user-friendly features like parameter sliders and live updates